

2023-2024
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People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
National Higher School of Mathematics



Resit exam (1^h30)

Name :

Groupe:

Exercise 1. We give in the following statistical table :

class	frequency
$[20, 30[$	100
$[30, 40[$	140
$[40, x_1[$	125
$[x_1, 70[$	200
$[70, 100[$	180
$[100, x_2[$	55

1. Given that the median of this distribution is equal to 56.8, calculate x_1 .

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2. Given that the arithmetic mean of the population studied is equal to 60.5, calculate the bound x_2 .

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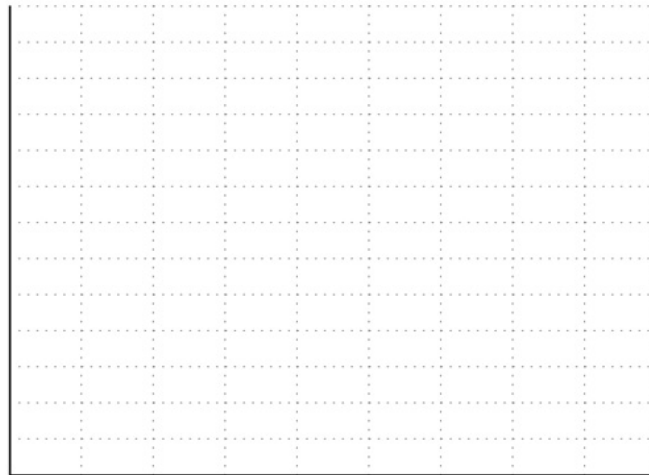
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- Plot the histogram of the distribution and calculate its mode.



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Exercice 2. The following table gives the distribution of 100 people questioned, according to two characteristics X and Y : X is the monthly income expressed in thousands of dinars and Y is the number of vacation departures during a given year

$X \backslash Y$	0	1	2	3	4
$[10, 15[$	4	4	6	1	0
$[15, 20[$	6	7	5	2	0
$[20, 25[$	4	4	6	5	1
$[25, 30[$	0	5	7	6	2
$[30, 35[$	1	8	6	5	5

- Give the number of people receiving a salary of at least 20 thousand dinars.
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- Give the number of people who have gone on vacation at least twice.
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- Among people earning between 20,000 and 25,000 dinars, what is the percentage of those who have gone on vacation at least once?
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4. Give the marginal distribution of X . Determine the mode and the first quartile.

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5. Determine the conditional distribution of Y given that $X \in [15, 20[$, then calculate the conditional mean of the number of vacation departures.

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6. define Z = "The holiday departure rate" as the ratio of the number of people who have gone on holiday at least once to the total number of people in the population considered.

(a) Complete the following table:

x_i	12.5	17.5			
z_i	0.11				

(b) Draw the scatter plot. What could we conclude?

[illegible][illegible]

(c) Calculate $Cov(X, Z)$ then $\rho(X, Z)$

[illegible]

(d) Write the regression equation of Z on X . What would be the vacation departure rate for people earning 37,500 DA?

[illegible]